2 Kalman Filtering

2.1 Linear Kalman Filtering

Most books on the Kalman filter are rather discouraging. [Str86]
—Gilbert Strang, 1986

In the previous chapter, we presented the fundamental equations of recursive least squares in (1.68)

$$P_i^{-1} = P_{i-1}^{-1} + A_i^T R_i^{-1} A_i$$

$$K_i = P_i A_i^T R_i^{-1}$$

$$\hat{x}_i = \hat{x}_{i-1} + K_i(y_i - A_i \hat{x}_{i-1})$$

Such equations, in essence, assume that the model of x is *static*

$$x_i = x_{i-1} \tag{2.1}$$

Kalman [Kal60] added dynamics to this static picture

$$\begin{aligned} x_i &= F_{i-1} x_{i-1} + G_{i-1} u_{i-1} + q_{i-1} \\ y_i &= A_i x_i + r_i \end{aligned} \tag{2.2}$$

where F_{i-1} is a matrix linearly combining the variables in the vector x_{i-1} to generate x_i, u_{i-1} will be a control vector (either a function of time, open loop, or of state x, closed loop) acted on by G_{i-1} (not necessarily the same size as F), q_{i-1} is random noise in the model, r_i is noise in the measurements y_i, and A_i is our matrix combining the variables in state x_i to produce measurement y_i.[1]

1. The q really involves more than random noise. Although there is a stochastic element to these process disturbances, q also represents the unknown and unmodelable aspects of the process. We will address this in much greater detail in chapter 8 on model inadequacy.

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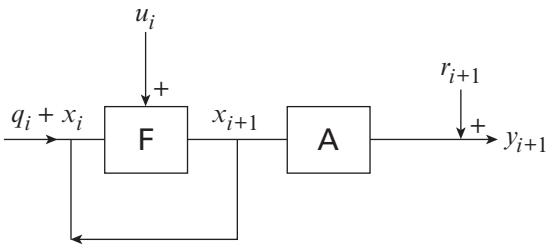


Figure 2.1
Schematic of Kalman filtering equations.

A frequently used schematic to illustrate such a system is shown in figure 2.1, which illustrates the dynamical mapping of x_i to x_{i+1} through F, the filtering through A, the addition of u to the dynamics, and the explicit addition of noise to dynamics, q, and measurement, r, where q_i and r_i are individual values drawn from Gaussian distributions with zero mean and variances Q and R, respectively

$$\begin{aligned} q_i &\sim N(0, Q_i) \\ r_i &\sim N(0, R_i) \end{aligned} \tag{2.3}$$

Let's propagate the mean of x. We will follow the logic and notation of Simon [Sim06a][2]

$$\begin{aligned} E(x_i) &= E(F_{i-1}x_{i-1} + G_{i-1}u_{i-1} + q_{i-1}) \\ &= F_{i-1}E(x_{i-1}) + E(G_{i-1}u_{i-1}) + E(q_{i-1}) \\ \bar{x}_i &= F_{i-1}\bar{x}_{i-1} + G_{i-1}u_{i-1} + 0 \qquad \{\text{treat } G_{i-1}u_{i-1} \text{ as const}\} \\ \bar{x}_i &= F_{i-1}\bar{x}_{i-1} + G_{i-1}u_{i-1} \end{aligned} \tag{2.4}$$

And now propagate the covariance P, the expectation of

$$\begin{aligned} (x_i - \bar{x}_i)(\cdots)^T &= (F_{i-1}x_{i-1} + G_{i-1}u_{i-1} + q_{i-1} - \bar{x}_i)\,(\cdots)^T \\ &= (F_{i-1}x_{i-1} + G_{i-1}u_{i-1} + q_{i-1} - F_{i-1}\bar{x}_{i-1} - G_{i-1}u_{i-1})\,(\cdots)^T \\ &= (F_{i-1}\,(x_{i-1} - \bar{x}_{i-1}) + q_{i-1})\,(\cdots)^T \\ &= F_{i-1}\,(x_{i-1} - \bar{x}_{i-1})\,(x_{i-1} - \bar{x}_{i-1})^T\,F_{i-1}^T + q_{i-1}q_{i-1}^T \\ &\quad + F_{i-1}\,(x_{i-1} - \bar{x}_{i-1})\,q_{i-1}^T + q_{i-1}\,(x_{i-1} - \bar{x}_{i-1})^T\,F_{i-1}^T \end{aligned} \tag{2.5}$$

2. Dan Simon's clear and self-contained book on Optimal State Estimation [Sim06a] is highly recommended as a complete and in-depth exposition on the details and logic of linear and nonlinear Kalman filtering.

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where the use of $(\cdots)$ indicates replication of the previous bracket contents. If we now take expectations, recalling that q_{i-1} is, by definition, uncorrelated with x_{i-1}

$$P_i = F_{i-1} P_{i-1} F_{i-1}^T + Q_{i-1} \tag{2.6}$$

Since we are working with the two time scales introduced in chapter 1, we again indicate the separation of time scales. A priori is the time, at $(k-1)^-$ or k^-, just before a new measurement, at $(k-1)^+$ or k^+, is made, producing an a posteriori value. Figure 2.2 schematizes this separation of time scales.

It is important to note now that in equation (2.4), the propagation of the mean $\bar{x}$ due to system dynamics, and the propagation of the covariance P in equation (2.5), both take place within the relatively long time scale from a posteriori to a priori in figure 2.2. We now recognize that the mean of x, $\bar{x}$, is now a moving target, the best estimate of x, $\hat{x}$ is

$$\hat{x}_i^- = F_{i-1} \hat{x}_{i-1}^+ + G_{i-1} u_{i-1} \tag{2.7}$$

and in the same spirit, a best estimate for the propagation of P is

$$P_i^- = F_{i-1} P_{i-1}^+ F_{i-1}^T + Q_{i-1} \tag{2.8}$$

Note, critically, why the variance of the noise on the dynamics of x drops out of the estimate of $\hat{x}$, but stays with the propagation of P. In equation (2.4), the expectation is taken over q_i, which is zero. In equations (2.5), the covariance produces a product of qq^T, whose expectation is the definition of variance Q. *The estimate of state error always worsens between measures. Wait long enough, and your uncertainty in a noisy process becomes global.*

This brings us to the relatively fast transition from a priori to a posteriori, that occurs after a new measurement is made (e.g., at time k^+ in figure 2.2). We first need to drag P_i^- across the threshold from a priori to a posteriori. And we return to our equations (1.68) for

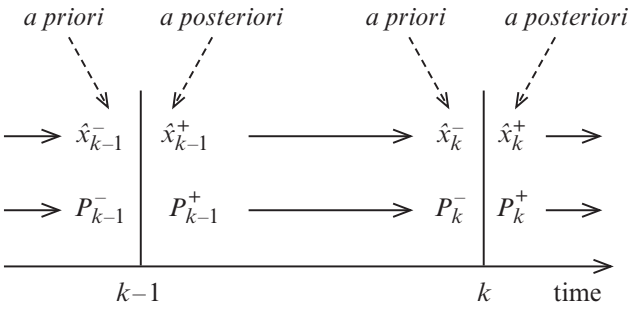


Figure 2.2
Separation of time scales. A priori is the time, at $(k-1)^-$ or k^-, just before a new measurement, at $(k-1)^+$ or k^+, is made, producing an a posteriori value. Modified after Simon [Sim06a] with permission.

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recursive least squares, which were intended for instantaneous use after measurements, to write[3]

$$P_i^+ = \left[(P_i^-)^{-1} + A_i^T R_i^{-1} A_i\right]^{-1} \tag{2.9}$$

and use this value of P_i^+ to calculate the next K_i

$$K_i = P_i^+ A_i^T R_i^{-1} \tag{2.10}$$

We use this K_i to bring $\hat{x}_i^-$ across the measurement threshold

$$\hat{x}_i^+ = \hat{x}_i^- + K_i(y_i - A_i \hat{x}_i^-) \tag{2.11}$$

Note that during the time without measurement, P_i^- was propagated in proportion to the dynamical noise covariance Q. This drives P_i larger. At the time of measurement, look at equation (2.9). A small measurement error covariance, R_i, will drive the inverse R_i^{-1} large. This tends to overwhelm $(P_i^-)^{-1}$, and the inverse of this sum will become small. In (2.10), a small P_i^+ will be countered by the large R_i^{-1}, and the sizable K_i will pay credence to the new measure y_i. Note that if R_i were huge relative to P_i, then the term $A_i^T R_i^{-1} A_i$ can be neglected, and in equation (2.10) K_i is driven small. Ignore y_i when the measure is unreliable. *When the situation is fluid, and the balance of uncertainties changes, Kalman's filter iteratively adjusts for these shifting priorities.*

To summarize the equations for linear Kalman filtering:

$$\begin{aligned}
x_i &= F_{i-1} x_{i-1} + G_{i-1} u_{i-1} + q_{i-1} \\
y_i &= A_i x_i + r_i \\
q_i &\sim N(0, Q_i) \\
r_i &\sim N(0, R_i) \\
\hat{x}_i^- &= F_{i-1} \hat{x}_{i-1}^+ + G_{i-1} u_{i-1} \\
P_i^- &= F_{i-1} P_{i-1}^+ F_{i-1}^T + Q_{i-1} \\
P_i^+ &= [(P_i^-)^{-1} + A_i^T R_i^{-1} A_i]^{-1} \\
K_i &= P_i^+ A_i^T R_i^{-1} \\
\hat{x}_i^+ &= \hat{x}_i^- + K_i(y_i - A_i \hat{x}_i^-)
\end{aligned} \tag{2.12}$$

3. There are a variety of ways of writing what are mathematically equivalent formulas for the Kalman filter. One often seen in the literature is $P_i^+ = (I - K_i A_i) P_i^-$, whose product can be less numerically stable [Sim06a] than the sum in (2.9). Later, we will introduce other formulations that are more convenient for the nonlinear ensemble methods of Kalman filtering.

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The balance between Q and R has some other considerations as well. From equation (2.6), P grows with propagation in proportion to the process (model) noise Q. Your intuition tells you that noise is bad, but intuition needs some more experience here. Note that if Q were very small, then P_i is minimized through the propagation step. In equation (2.9), a small P_i^- generates a term that tends to overwhelm the other term when the inverse $(P_i^-)^{-1}$ is taken. The final inverse in equation (2.9) makes P_i^+ small, which in (2.10) drives K_i small. Over time, the iterations of very small process noise tend to drive P small, and the filter stops paying attention to new measurements. So adding additional uncertainty Q to the filter can actually improve its performance. There is something deeper here—it involves what we will later term *covariance inflation*, as well as the broader issue of *model inadequacy*, to which chapter 8 is devoted. But for now, use enough Q.

If the Kalman filter in equations (2.12) is an accurate one, without significant modeling errors, the innovations $(y_i - A_i\hat{x}_i^-)$ become a Gaussian random variable. These *residuals* are what is left after extracting the maximal amount of information from a signal. *The Kalman filter really is a filter*. It extracts information out of a noisy signal.

One unfortunate aspect of the original Kalman filter equations (2.12) is that they describe linear systems.[4] It is unfortunate because there are so few close-to-linear systems in biology.[5] And I'll just be blunt in saying that there are no linear neuronal dynamics. Nevertheless, the power of linear Kalman filtering has intrigued neuroscientists. In a technical report, Bousquet and colleagues laid out a linear Kalman filter framework to account for hippocampal probabilistic spatial localization [BBH98]. Just as in spatial learning in the hippocampus, there appears to be explicit predictive coding in the visual system, and extended (linearized) Kalman filtering has been placed in this framework [RB97]. In addition, the issue of content-addressable memory and Kalman filtering was discussed by Ballard [Bal97]. Limitations of these linear approaches were discussed in some detail by Eliasmith and Anderson [EA03]. While these researchers in the 1990s were struggling with linear Kalman frameworks, the robotics and numerical weather prediction communities were developing explicit nonlinear adaptations of the Kalman filter that were computationally tractable. These newer methods provided the motivation for this book.

2.2 Nonlinear Kalman Filtering

Unfortunately, linear systems do not exist. . .
—Dan Simon, 2006 [Sim06b]

4. If you are engineering a man-made system, then you can constrain the nonlinearities and let Kalman filtering revolutionize your control capabilities.

5. A thoughtful discussion of control of *bilinear systems* in biology can be found in [Moh91a, Moh91b].

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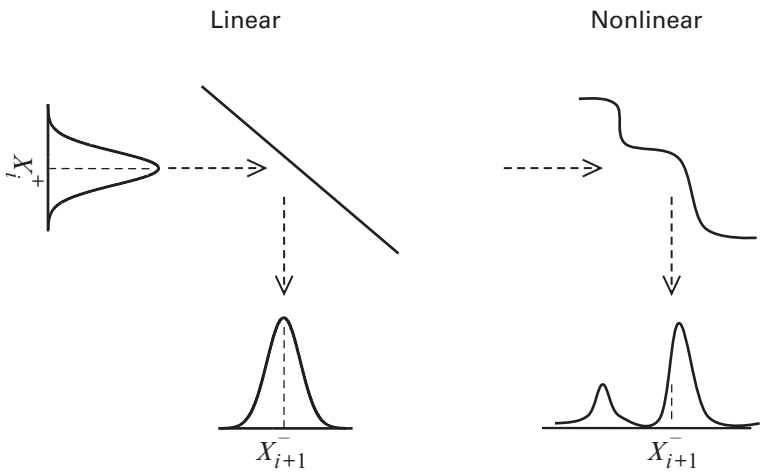


Figure 2.3
Comparison of the propagation of the mean and uncertainty of $\hat{x}_i^+$ to $\hat{x}_{i+1}^-$. This propagation corresponds to the time from a posteriori to a priori in figure 2.2. The black dashed lines within the distributions indicate the true mean values of the state x. Modified after Thrun et al. [TBF05].

As illustrated in figure 2.2, Kalman filtering has two parts: a propagation of state and uncertainty, and a faster Bayesian fusion of state prediction with data. Bayes' rule is immutable for our purposes. The propagation is not.

Thrun and colleagues [TBF05] provide a detailed analysis of comparative propagation schemes. We will follow their logic here. First, in linear Kalman filtering, one passes the mean (your best estimate of state $\hat{x}$) and uncertainty from Gaussian distributions through linear transformations—what comes out of linear transformation of a Gaussian distribution is another Gaussian (figure 2.3).

Propagating the same Gaussian distribution through a nonlinear function distorts it (see figure 2.2). If you had an analytic solution to this propagation, and the subsequent propagation of the no-longer-Gaussian distributions that result, you could write the general nonlinear version of Kalman's filter and use it. But there are no such general solutions to this problem.

The extended Kalman filter (EKF) is where one linearizes a nonlinear function at a particular point. Literally, draw a straight-line tangent to this point, and convert the nonlinear propagation problem into the linear one. Figure 2.4 illustrates the linearized solution.

For a wide range of problems this works very well. But in many circumstances, there is no adequate linearization that will capture the propagation of mean and uncertainty with reasonable fidelity to the underlying problem. Figure 2.4 illustrates why there can often be significant error in the linearized propagation of the mean $\hat{x}_{i+1}^-$.

Although the EKF (in its many forms) is a widely used filtering strategy, over thirty years of experience with it has led to a general consensus within the tracking and control community that it is difficult to

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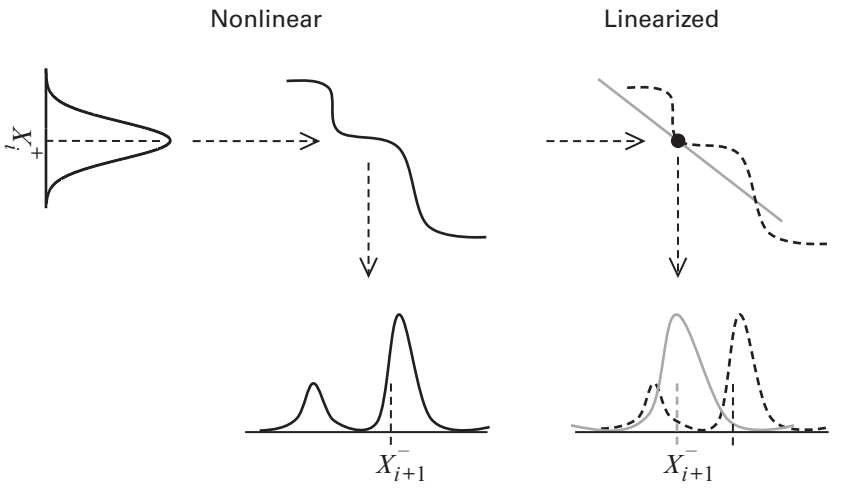


Figure 2.4
Comparison of nonlinear versus linearized propagation of the mean and uncertainty of $\hat{x}_i^+$ to $\hat{x}_{i+1}^-$. There is significant distortion of the mean (gray dashed line) from the true mean (black dashed line). Modified after Thrun et al. [TBF05].

implement, difficult to tune, and only reliable for systems which are almost linear on the time scale of the update intervals. [JU97a]

In range finding, one might wish to convert sines and cosines from individual sonar or radar systems to a common Cartesian coordinate. Linearization gives inaccurate results. This simple yet intractable problem motivated the push toward explicitly nonlinear methods [JU97b], [JU97a].[6]

The most accurate way to numerically solve the nonlinear propagation problem is to approximate the distribution by many individual states, and solve the passage of each state through the nonlinear propagation equations. Such methods fall under the general rubric of Monte Carlo methods, although in control engineering the terminology is generally that of particle filters. If you use enough particles (generally picked at random), you can faithfully approximate any distribution, and then process each particle one-by-one as shown in figure 2.5. The mean value of x is more faithfully propagated in this illustration (left side of figure 2.5). Nevertheless, the computational load of such Monte Carlo methods can be high, and this often prohibits real-time use.

In 1997, Julier and Uhlmann published two proceedings papers. In the first [JU97a], they demonstrated that by parameterizing a Gaussian distribution (such distributions are fully characterized by their means and standard deviations, as in equation (2.3)), that they could

6. A lucid and expanded discussion of Julier and Uhmann's more difficult discussion can be found in [Sim06a]. Note that were a Taylor's series expansion used to keep the second order terms of the sines and cosines, the inaccuracies of this simple range-finding problem would be better addressed. Nevertheless, many of the problems in this book would not be well served by higher-order attempts at expansion.

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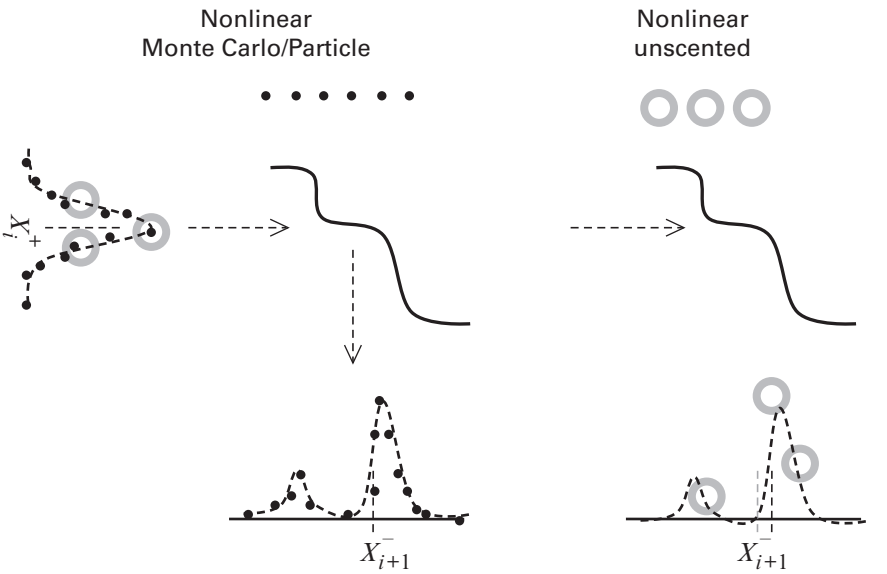


Figure 2.5
Comparison of Monte Carlo (particle) propagation versus unscented Kalman filter using the parameterization of the mean and uncertainty of $\hat{x}_i^+$ by individual *sigma* points. There is less distortion of the mean (gray dashed line) from the true mean (black dashed line) in comparison with linearization (cf. figure 2.4). Modified after Thrun et al. [TBF05].

dramatically cut down on the number of particles required, yet preserve accuracy. They called their parameterized points *sigma points*. Their method was "founded on the intuition that it is easier to approximate a Gaussian distribution than it is to approximate an arbitrary nonlinear function or transformation" [JU97a].

Compared with the extended Kalman filter, their *unscented* Kalman filter required no computation of derivatives in linearizing nonlinear systems (Jacobians in higher dimensions), and proved more accurate. There is no truncation of the nonlinear function from linearization (recall that linearization is a truncation of the Taylor's series expansion of the nonlinear function at a particular point). And they preserved the statistics of the moments of the distribution they were propagating to second order (mean and variance). Figure 2.5 illustrates this method in comparison with a full Monte Carlo simulation.

Significantly, in their second 1997 paper [JU97b], less well read by the community, Julier and Uhlmann pointed out that one can add more points to the parameterization, such as kurtosis of the distribution, to bring the accuracy of the representation to fourth order. This substantially improved the accuracy of their range finding position calculations.[7]

7. It is a remarkable testament to the peer review system that the peer review of the findings in these two papers would take four submissions over six years. They graciously wrote: "The authors would like to thank ... the referees for their many constructive remarks" [JUDW00].

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Sigma points are generated deterministically. In the second order unscented Kalman filter, for an n-dimensional state variable x, there is an n-dimensional uncertainty distribution. The $2n+1$ sigma points, X, are generated as

$$\begin{aligned} X_0 &= \bar{x} \\ X_i &= \bar{x} + \left(\sqrt{(n+\kappa)P}\right)_i \\ X_{i+n} &= \bar{x} - \left(\sqrt{(n+\kappa)P}\right)_i \end{aligned} \tag{2.13}$$

where $i = 1 \ldots n$ and $\bar{\cdot}$ indicates average. The mean of the distribution is chosen as the first sigma point $\bar{x}$. P is the covariance matrix for the errors in x. The square root is the matrix square root, whose computation is greatly abetted by the availability of efficient decompositions such as Cholesky's.[8] The matrix square root of M is defined as

$$M \equiv \left(\sqrt{M}\right)^T \sqrt{M}$$

and in practice, one chooses the ith row (or column) from $\sqrt{M}$ to sum with the average of $\bar{x}$ in equation (2.13).

As in other forms of particle filtering, one weights the sigma points. Julier and Uhlman used for their weights, q,

$$\begin{aligned} q_0 &= \kappa/(n+\kappa) \\ q_i &= 1/2(n+\kappa) \\ q_{i+n} &= 1/2(n+\kappa) \end{aligned} \tag{2.14}$$

but in practice, one can often drop the q_0 weight (it is only one point), and then the weights are all equal and can be ignored. Later in the chapter, we will just use $2n$ sigma points.

Kappa (κ) is fascinating. Julier and Uhlman used this as an extra degree of freedom to "fine tune" the approximation and reduce errors [JU97a]. What κ does in practice is *increase* the size of the uncertainty "ball" that the sigma points represent. In recent years, numerical meteorology has investigated this type of *covariance inflation* in considerable detail [AA99], [HKS07], and we will deal with this in more detail later.

In higher dimensions, the unscented Kalman filter propagation appears as in figure 2.6.

The propagation of sigma points provides a ready route to calculate the mean and covariation of the propagated points in order to realize the new a priori values of $\hat{x}_i^-$ and P_i^-. But Julier and Uhlmann [JU97a] used a different formulation for calculating P_i^+ and K when using the unscented Kalman filter to make it computationally simpler.

Recall the formula (2.11) for the final update of $\hat{x}_i^+$

8. Cholesky's work on this decomposition far predated the advent of digital computers.

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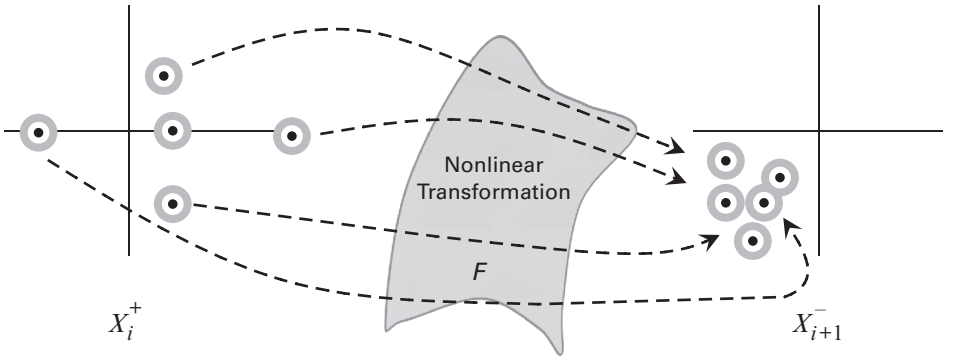


Figure 2.6
Schematic of unscented Kalman filter propagation of sigma points (circles). The transformed points are used to calculate the new a priori estimate of x and its covariance P. Modified after [JU97a].

$$\hat{x}_i^+ = \hat{x}_i^- + K_i(y_i - A_i\hat{x}_i^-)$$

One way to view K_i is as a "derivative" that evaluates the rate of change of x in relation to y

$$K_i = \frac{\hat{x}_i^+ - \hat{x}_i^-}{y_i - \hat{y}_i^-} \tag{2.15}$$

Using this notion of linearity in K, a *best linear unbiased estimate* for K can be alternatively derived by setting $z = x - \hat{x}_i$, and noting that the covariance of z, P_z, is

$$P_z = E\left[(x - \hat{x} - (\bar{x} - \bar{\hat{x}}))(\cdots)^T\right] \tag{2.16}$$

Minimizing the trace of the matrix P_z minimizes the sum of the variances of the individual components of x, and this will generate the best K. A somewhat intricate derivation in Simon [Sim06a] demonstrates that this trace is minimized when

$$K = P_{xy}P_{yy}^{-1} \tag{2.17}$$

and further consideration [Sim06a] yields

$$P_{xx}^+ = P_{xx}^- - K\,P_{xy}^T \tag{2.18}$$

This best linear statistical derivation makes use of some new notation on the covariances

$$\begin{aligned} P_{xx} &= E\left[(x - \hat{x})(x - \hat{x})^T\right] \\ P_{xy} &= E\left[(x - \hat{x})(y - \bar{y})^T\right] \\ P_{yy} &= E\left[(y - \bar{y})(y - \bar{y})^T\right] \end{aligned} \tag{2.19}$$

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Despite the very different appearance of these equations (2.17) through (2.19) from (2.9) through (2.11), they are mathematically equivalent [Sim06a]. We will now see why this formulation is so transparent for the unscented Kalman filter.

We now write our equations (2.2) as

$$\begin{aligned} x_i &= F_i(x_{i-1}, u_{i-1}) + q_{i-1} \\ y_i &= A_i(x_i) + r_i \end{aligned} \tag{2.20}$$

where F, G, and A are no longer just matrices multiplying x and u, but are now possibly nonlinear functions $F(\cdot)$ and $A(\cdot)$ of x and u.

In figure 2.7, we show a schematic of the full propagation of values through the unscented Kalman filter. Sigma points are generated from the covariance P_{xx}^{+} of $\hat{x}_i^{+}$, which are then propagated through the dynamical equation $F(\hat{x}_i^{+}) + q_i$ from equation (2.20). This generates a new ensemble of points whose average is the new $\hat{x}_{i+1}^{-}$. One can read the propagated P_{xx}^{-} off of the ensemble of points by taking their covariance with respect to the average $\hat{x}_{i+1}^{-}$. Then you pass these points through the observation function $\hat{y}_{i+1} = A_{i+1}(\hat{x}_{i+1}) + r_{i+1}$, which lets you estimate both the expected $\hat{y}$, and the covariance in the expected $\hat{y}$, P_{yy}.

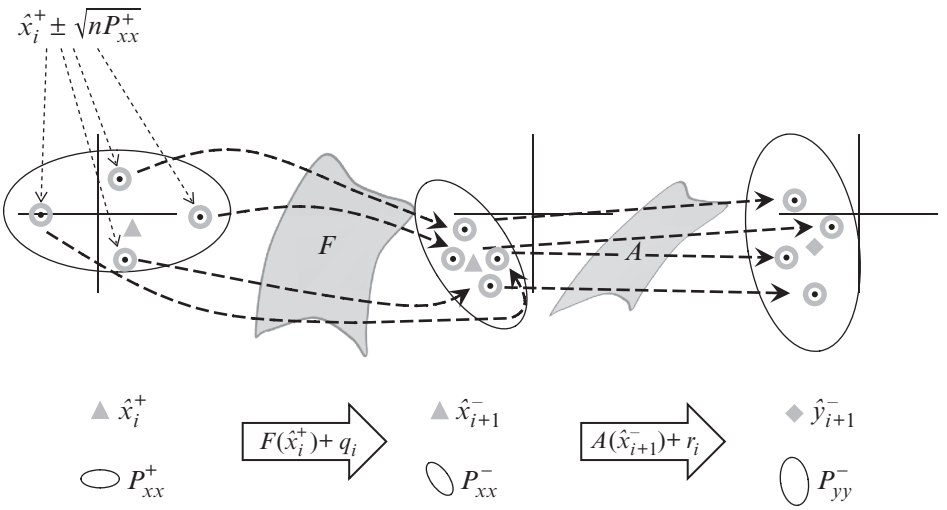


Figure 2.7
Schematic of the propagation components of the unscented Kalman filter. The notation in this figure is matched to the notation in the summary equations (2.22). At the left, using the present estimate of $\hat{x}_i^{+}$, the weighted matrix square root of the covariance P_{xx}^{+} is used to generate sigma points (circles). These sigma points are iterated forward using the nonlinear equations F. The mean of the new ensemble is $\hat{x}_{i+1}^{-}$, and the new covariance P_{xx}^{-} generated from the ensemble about this mean. The points are further transformed by the observation function A, to generate the estimate of the next measurement, $\hat{y}_{i+1}^{-}$, and calculate P_{yy}^{-}. All of this takes place *before* the new measurement y is taken, after which K is used to propagate P_{xx}^{-} and $\hat{x}_{i+1}^{-}$ to P_{xx}^{+} and $\hat{x}_{i+1}^{+}$.

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Using P_{xy} calculated as in (2.19), one then calculates K from (2.17), and finally everything is in order to calculate P_{xx}^{+} from (2.18). The final update of x is now written as

$$\hat{x}_{i+1}^{+} = \hat{x}_{i+1}^{-} + K_{i+1}(y_{i+1} - \hat{y}_{i+1}^{-}) \tag{2.21}$$

Let's summarize the equations for the unscented Kalman filter using $2n$ sigma points:

$$\begin{aligned}
x_i &= F_i(x_{i-1}, u_{i-1}) + q_{i-1} \\
y_i &= A_i(x_i) + r_i \\
q_i &\sim N(0, Q_i) \\
r_i &\sim N(0, R_i)
\end{aligned}$$

$$\begin{aligned}
X_j &= \hat{x}_i^{+} \pm \left(\sqrt{n P_{xx}}\right)_j, \quad j = 1 \ldots 2n \\
\tilde{X}_j &= F(X_j) \\
\tilde{Y}_j &= A(\tilde{X}_j)
\end{aligned}$$

$$\begin{aligned}
\hat{x}_{i+1}^{-} &= \frac{1}{2n}\sum_{j=1}^{2n} \tilde{X}_j \\
\hat{y}_{i+1}^{-} &= \frac{1}{2n}\sum_{j=1}^{2n} \tilde{Y}_j
\end{aligned} \tag{2.22}$$

$$\begin{aligned}
P_{xx}^{-} &= \frac{1}{2n}\sum_{j=1}^{2n} (\tilde{X}_j - \hat{x}_{i+1}^{-})(\tilde{X}_j - \hat{x}_{i+1}^{-})^T + Q_i \\
P_{xy} &= \frac{1}{2n}\sum_{j=1}^{2n} (\tilde{X}_j - \hat{x}_{i+1}^{-})(\tilde{Y}_j - \hat{y}_{i+1}^{-})^T \\
P_{yy} &= \frac{1}{2n}\sum_{j=1}^{2n} (\tilde{Y}_j - \hat{y}_{i+1}^{-})(\tilde{Y}_j - \hat{y}_{i+1}^{-})^T + R_{i+1}
\end{aligned}$$

$$\begin{aligned}
K &= P_{xy} P_{yy}^{-1} \\
P_{xx}^{+} &= P_{xx}^{-} - K P_{xy}^{T} \\
\hat{x}_{i+1}^{+} &= \hat{x}_{i+1}^{-} + K(y_{i+1} - \hat{y}_{i+1}^{-})
\end{aligned}$$

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A few comments are in order for what appears complex. The notation here is unified with figure 2.7. We have indexed the sigma points X and Y by j, which reflects that, for dimension n (the number of elements in the state vector x), we have $2n$ sigma points. Don't confuse j with the time indices i. The difference between Kalman's original filter and the unscented Kalman filter is all in the ensemble updating for the propagation steps. These propagations bring us to the a priori condition for the final update when the new measurement y_{i+1} arrives.

I list in equations (2.22) adding the process noise Q to the estimation of P_{xx}, and observation noise R to the estimation of P_{yy}. There are many variants in how to handle adding these quantities, and we will deal with many of these intricacies later in the book. For now, I have dropped the extra parameter κ that Julier and Uhlmann used in equations (2.13)—but it is all the same issue. Embracing uncertainty, whether random noise, or model inadequacy, is a major topic to be dealt with.

The last three equations of (2.22) are identical to the original Kalman filter. Bayes' rule still rules the *assimilation* of the data point. On the other hand, there is no longer any need to be concerned with approximating the nonlinear functions F and A by something inaccurate such as a linearization. The sigma points are each numerically passed one-by-one through these intact nonlinear equations, regardless of how unpleasant or difficult to solve analytically such equations might be. The only issue remaining is how to numerically integrate those equations, and we will deal with this later in the book.

While Julier and Uhlmann were working out their method, they were not alone. One of the strange facts of academic life, is that we can spend a lifetime in one building, and never meet, discuss, or perish the thought read the literature of our colleagues in another department. In 1994, Evensen [Eve94] published an article seeking to adapt the Kalman filter to the tracking and prediction of ocean dynamics. His ensemble Kalman filter, has had a profound impact on numerical meteorology and oceanography (see, e.g., [EvL00], [OHS+04], [BHK+06], [HKS07]). In the geophysical literature, ensemble Kalman filtering is the more commonly used term, whereas in robotics, the term unscented Kalman filter is more common. The particular method of deterministically choosing the sigma points is associated with unscented Kalman filtering. But this is a fascinating example of parallel development of scientific ideas.

2.3 Why Not Neuroscience?

It is interesting to consider why the techniques discussed this chapter that have revolutionized robotics and weather prediction have not had a significant impact on neuroscience. There are several good reasons for this.

First, the subject matter, despite my attempts at an approachable exposition, is unfriendly. No neuroscientist or physician has in the past come to their profession with a strong grounding in the appropriate background.

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Second, the older theory, linear and linearized approximations to dynamics, was not broadly applicable to the nonlinearities of neuronal dynamics.

Third, we needed neuronal models suitable to the tasks we would ask of them. Computational neuroscience models have been maturing for nearly as long as modern control theory—both had their origins in the 1950s. For the past several decades, the fidelity of these neural models has grown ripe for fusing with control theory.

Fourth, we needed suitable nonlinear control algorithms that were suitable for nonlinear dynamics. Once Evensen, or Julier and Uhlmann, had done their work, it needed to percolate through and across disciplinary boundaries and creep into neuroscience settings. The first point of entry seems clear. In a relatively obscure physics journal, Voss, Timmer, and Kurths [VTK04] made the leap in 2004.

And finally, one might ask why I chose to focus on the unscented Kalman filter for this introduction? Because this is what Voss and colleagues first used in their seminal paper bridging from nonlinear control theory to neuronal dynamics. Their algorithmic framework has subsequently proven instrumental in applications to several neuronal systems that we will discuss in detail in this book. Nevertheless, a considerable amount of cutting-edge research in ensemble tracking and prediction of nonlinear systems is coming out of meteorology, and we will cross over between these literature streams frequently.

But before we can get to the results of Voss et al. [VTK04], we must first discuss the Hodgkin-Huxley equations in the next chapter, and then the reduction of Hodgkin-Huxley dynamics in the Fitzhugh-Nagumo equations in the subsequent chapter, before the application of equations (2.22) will be transparent.

Exercises

2.1. For the first (and likely last) time in your life, let's work the steps of the Kalman filters by hand. Return to the simple example from equation (1.20)

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

where we calculated from least squares in (1.27) that $\hat{x} = 2$, and from (1.54) that $P = \sigma^2/2$, and let's use the linear Kalman filter equations from (2.12).

Let's assume a static model, where the dynamics are trivial

$$F = I, \text{ so that } F_{i+1} = F_i$$

and let's assume again that the variable x is directly observable, so that

$$A = I, \text{ and } A_{i+1} = A_i$$

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We will assume that the uncertainty in both model, F, and observation, A, are both equal to σ^2. The initial conditions are therefore

$y_1 = 1$

$\hat{x}_1^+ = y_1$

- Calculate $\hat{x}_2^-$
- What is P_1^+?
- What is P_2^-?
- What is P_2^+?
- What is K_2?
- Calculate $\hat{x}_2^+$.

Does your answer differ from the least squares solution to this problem? It should.

2.2. To clarify the results in exercise 2.1, reverse the order of the measurements, so that $y_1 = 3$ and $y_2 = 1$. Calculate $\hat{x}_2^+$.

Hopefully your answer again differs from the least square solution, and also differs from the result in exercise 2.1. Kalman filtering tends to (exponentially) *forget* past data, and weights the most recent data more heavily.

2.3. Let's now calculate the same results for the static model in exercise 2.1, but now use the unscented Kalman filter using equations (2.22). Assume that the variance P_{xx} to start with is 1, and that for this simple one-dimensional problem, $n = 1$. We will start again setting $\hat{x}_1 = y - 1 = 1$, and the two sigma points will simply be 2 and 0.

- What are P_{xx}^-, P_{xy}, and P_{yy}?
- What is K?
- Calculate $\hat{x}_2^+$.

[Hint: The answer should be the same as the least squares solution.]

- Now calculate P_{xx}^+.
- Assuming that you were running your unscented filter for awhile, the newest P_{xx} would be the P_{xx} for calculating the next set of sigma points. So repeat the preceding calculation for $y_1 = 1$ and $y_2 = 3$, except start by using the last P_{xx} you had calculated above to generate the sigma points. What is $\hat{x}_2^+$ now? It should now be very close to the value from the linear Kalman filtering in 2.1, weighted in favor of the most recent measurement. Sometimes you just have to let a dynamical system settle down from its initial transient response. You should think of Kalman filters as dynamical systems.

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